Boston College || Woods College of Advancing Studies || ADAN|ADEC7900 Software Tools

R Homework 1

Objective

Your objective is to demonstrate basic R programming skills by creating and testing your own functions. By the end of the assignment, your program should handle both valid and erroneous inputs. You will post your R program to a GitHub repository where the instructor will view and run your work.

Instructions

1. You should submit this assignment as a single R script to a GitHub repository.
    - Important: your entire code should run without any errors! Debugging errors is a valid use of AI in this course.
2. Start with the provided template.
3. `#Comment your work so readers understand your steps.`
4. Use only base R functions. Do not use functions from external packages.
5. Remove all non-essential code and `#personal notes` once you complete debugging.

FizzBuzz

In this question you will need to replicate a common group word game for children called "FizzBuzz." Every computer science student loves FizzBuzz.

1. Define a function `fizz(n)` that takes a whole number n and returns a character string "Fizz" if n is divisible by 3 and an empty string `""` if it is not.

Use `%%` operator to calculate a remainder from a division, e.g. `7 %% 5`

Hint: Avoid using `print()` to display a result. Instead, use return() . They look similar, but returning a variable to the R environment allows you to save it as a global variable or input it to another function for further processing.

2. Define a similar function `buzz(n)` that takes a whole number n and returns a character string `"Buzz"` if n is divisible by 5 and an empty string `""` if it is not.

3. Use the two functions above to define a function `fizzbuzz(n)` that takes a whole number n and returns one of the following results:

- a character string `"Fizz"` if n is divisible by 3
- a character string `"Buzz"` if n is divisible by 5
- a character string `"FizzBuzz"` if n is divisible by both 3 and 5
- an empty character string `""` if none of the above

There should be no conditional statements inside `fizzbuzz(n)` checking for remainder of

the division (e.g. if (`n %% 5 == 0`)). Instead, you should call your previously defined functions `fizz(n)` and `buzz(n)` from inside `fizzbuzz(n)` to evaluate the supplied whole number argument. You can still use conditional statements to check for other things, such as whether the

output of `fizz(n)` is equal to `"Fizz"` or not.

Hint: the whole `fizzbuzz(n)` code can be written as one simple line using paste()

function to combine the outputs of `fizz()` and `buzz()` .

4. Use the function `fizzbuzz()` to define a function `fbr(n,m)` that takes two whole numbers n and m as two inputs, goes through all whole numbers from n up to m, returning a vector that for each number between n and m contains either a string `"Fizz/Buzz/FizzBuzz"` as defined by `fizzbuzz(n)` or the number itself.

Example: `fbr(10,15)` should return the following vector:

`("Buzz","11","Fizz","13","14","FizzBuzz")`

4.1 Make `fbr(n,m)` check whether supplied arguments are integer variable type (using `is.integer()` function) and print an error message (using `stop()` function) otherwise.

4.2 Make `fbr(n,m)` check whether n is less or equal to m and print

an error message (using `stop()` function) otherwise.